

ABSOLUTE CONVERGENCE & CONDITIONAL CONVERGENCE

Motivation:

Lesson 31

In previous lessons we have been dealing with

$$\sum_{n=1}^{\infty} U_n, \text{ where } U_n \text{ - are positive terms.}$$

for such series converge/diverge is determined by ratio test, Comparison test or test 1.

There are other series $\sum_{n=1}^{\infty} U_n = U_1 - U_2 + U_3 - U_4 + U_5 - U_6 + \dots$ called alternating series, the sign of U_n term is $+/-$ alternates.

Defn: If the series $\sum_{n=1}^{\infty} |U_n|$ converges, the series $\sum_{n=1}^{\infty} U_n$ alternating series is said to be absolutely convergent.

Defn: If the series (alternating series) $\sum_{n=1}^{\infty} U_n$ is convergent and $\sum_{n=1}^{\infty} |U_n|$ is divergent, then we say the series $\sum_{n=1}^{\infty} U_n$ is Conditionally Convergent.

Example 1

The series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \dots$ is convergent

but $\sum_{n=1}^{\infty} |U_n| = \sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \dots$ is divergent

therefore the series $\sum_{n=1}^{\infty} U_n = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ is Conditionally Convergent.

Example 2

Find the range of values of x for which the following series is absolutely convergent

$$\frac{x}{2 \times 5} - \frac{x^2}{3 \times 5^2} + \frac{x^3}{4 \times 5^3} - \frac{x^4}{5 \times 5^4} + \frac{x^5}{6 \times 5^5} - \dots$$

Solution

The series is $\sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{(n+1) 5^n}$

Here $|u_n| = \frac{|x|^n}{(n+1) 5^n}$ and $|u_{n+1}| = \frac{|x|^{n+1}}{(n+2) 5^{n+1}}$

$$\left| \frac{u_{n+1}}{u_n} \right| = \frac{|x|^{n+1}}{(n+2) 5^{n+1}} \cdot \frac{(n+1) 5^n}{|x|^n} = \frac{|x|}{5} \left(\frac{n+1}{n+2} \right) = \frac{|x|}{5} \left(\frac{1 + \frac{1}{n}}{1 + \frac{2}{n}} \right)$$

$$\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \frac{|x|}{5}$$

For absolute convergence, $\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| < 1$

ie $\frac{|x|}{5} < 1$

$$|x| < 5$$

$-5 < x < 5$, Range of x for absolute convergence.